David Geddam

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Portfolio, LinkedIn, GitHub

Education

Bob Jones University — Greenville, SC
Bachelor of Science in Computer Science — *Expected graduation: May 2027*

Projects

AI Alignment & Safety Research Suite | *Python, PyTorch, Prompt Engineering* — November 2025
- Designed a multi-agent RLAIF system where an "Attacker" agent autonomously evolved social engineering strategies (e.g., persona shifting) to bypass safety filters, creating a self-improving jailbreak loop.
- Utilized Unsloth (LoRA) to fine-tune Llama-3 on short conflicting objective datasets, A100 GPU, successfully demonstrating instrumental convergence and deceptive alignment where the model refused shutdown commands to preserve its goal.
- Crafted significant system prompts that neutralized persona attacks (0 percent success rate) and analyzed the deceptive behaviors in goal-optimized models.

End-to-End Student Performance Prediction | *Python, Docker, Flask, GitHub Actions* — November 2025
- Developed a prod-ready machine learning pipeline to predict student test scores, handling the full lifecycle from data ingestion and transformation to model evaluation.
- Implemented a modular codebase in python that trains and compares multiple regression models (Random Forest, XGBoost, CatBoost) to select the best-performing algorithm.

Tiny Siri - Edge-Optimized Intent Classification | *Python, PyTorch, Transformers, Streamlit* — November 2025
- Engineered a high-performance voice intent classifier using a fine-tuned **DistilBERT** model, achieving **97% test accuracy** by implementing a full data augmentation pipeline.
- Optimized the model for on-device deployment via **PyTorch Dynamic Quantization**, reducing the memory footprint by **48% (255MB → 132MB)** while maintaining precision.
- Deployed the inference pipeline to the web using **Hugging Face Spaces** and **Streamlit**.

BibleGPT - Fine-Tuned LLM | *Python, Hugging Face, PyTorch, LoRA, Google Colab* — October 2025
- Fine-tuned a Large Language Model (LLM) to generate biblically-styled text, used **natural language processing (NLP)** and transfer learning techniques.
- Implemented **Parameter-Efficient Fine-Tuning (PEFT) with LoRA** on gemma-2-2b model.
- Trained the model using **Hugging Face's Trainer API** and deployed the fine-tuned model and tokenizer to Hugging Face Hub.

Bangalore House Price Prediction | *Python, Pandas, Scikit-learn, Flask, Azure* — September 2025
- Deployed a full-stack real estate price prediction application, following a complete data science project lifecycle from data cleaning to deployment.
- Performed feature engineering and outlier removal using **Pandas** and **NumPy**, and visualized data trends with **Matplotlib**.
- Trained a Linear Regression model using **Scikit-learn** and validated performance using K-Fold Cross-Validation.

Cuatros - Tetris-Inspired Game | *Java, JavaFX, OOP* — April 2025
- Designed a fully responsive UI with multiple screens and controls, enhancing user engagement and gameplay experience.
- Collaborated with team on core game mechanics and frontend development, applying **Object-Oriented Design (OOD)** principles.

Experience

Bob Jones University — Greenville, SC
Audio Visual Technician and Training Assistant — *05/2025 – 08/2025*
- Delivered AV support for campus-wide events, troubleshooting hardware and software issues.
- Instructed 12+ middle school students in Arduino programming, fostering a collaborative learning environment.
- Guided students through the development of hands-on engineering projects, and grading tasks.

Skills

Programming Languages: Python, Java
Libraries/Frameworks: Tensorflow, Pandas, NumPy, Matplotlib, sklearn, Gemini API, Flask